

Three-dimensional distribution of amino groups improves perovskite crystallization and defect passivation in high-performance photovoltaics

Received: 17 March 2026

Accepted: 16 July 2026

Published online: 25 July 2026

Cite this article as: Jiang, X., Yang, G., Zhang, B. *et al.* Three-dimensional distribution of amino groups improves perovskite crystallization and defect passivation in high-performance photovoltaics. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-75990-x>

Xiaoqing Jiang, Guangyue Yang, Bingqian Zhang, Yu Lei, Panyu Wang, Na Shi, Kaiwen Dong, Likai Zheng, Yue Qiang, Shiwei Liu, Zhongjin Shen, Marina Freitag, Chongwen Li, Shuping Pang, Mohammad khaja Nazeeruddin & Xin Guo

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Three-dimensional distribution of amino groups improves perovskite crystallization and defect passivation in high-performance photovoltaics

Xiaoqing Jiang,^{#1,2} Guangyue Yang,^{#1,2} Bingqian Zhang,^{#1} Yu Lei,² Panyu Wang,² Na Shi,² Kaiwen Dong,³ Likai Zheng,⁴ Yue Qiang,⁵ Shiwei Liu,² Zhongjin Shen,^{6*} Marina Freitag,⁶ Chongwen Li,^{3*} Shuping Pang,¹ Mohammad khaja Nazeeruddin^{7*} and Xin Guo^{8*}

¹ State Key Laboratory of Photoelectric Conversion and Utilization of Solar Energy, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao New Energy Shandong Laboratory, Shandong Energy Institute, Qingdao 266101, China.

² College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China.

³ School of Electronic Science and Engineering, Southeast University, Nanjing, China.

⁴ Laboratory of Photonics and Interfaces, École Polytechnique Fédérale de Lausanne, CH-1015 Lausanne, Switzerland.

⁵ Engineering Technology Research Institute, PetroChina Xinjiang Oilfield Company, Karamay 834000, China.

⁶ School of Natural and Environmental Sciences, Newcastle University, NE1 7RU, Newcastle upon Tyne, United Kingdom.

⁷ School of Integrated Circuits, Southeast University, Wuxi, Jiangsu, 214026, China.

⁸ State Key Laboratory of Photoelectric Conversion and Utilization of Solar Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China.

[#] X.J., G.Y. and B.Z. contributed equally to this work.

* Corresponding authors: zhongjin.shen@newcastle.ac.uk (Z.S.); chongwen.li@seu.edu.cn (C.L.); 108109144@seu.edu.cn (M.K.N.); guoxin@dicp.ac.cn (X.G.)

Abstract

Incorporating organic molecules with diverse functional groups to improve film quality has emerged as a crucial strategy for realizing high-performance perovskite solar cells (PSCs). Nevertheless, the role of spatial distribution of those functional groups in governing passivation efficacy and perovskite crystallization remains insufficiently investigated. Here, we introduce three amino-containing molecules, bis(4-aminophenyl)methane (2APM), tris(4-aminophenyl)methane (3APM) and tetrakis(4-aminophenyl)methane (4APM), featuring distinct spatial distributions of amino groups, into the perovskite precursor solution as in-situ regulators. Among them, 4APM exhibits the strongest interactions with PbI_2 and formamidinium iodide (FAI) by virtue of its three-dimensional (3D) spatially distributed amino groups, most effectively suppressing uncoordinated Pb^{2+} defects and enhancing perovskite film crystallinity. As a result, PSCs incorporating 4APM achieve a stabilized power conversion efficiency (PCE) of 26.26%, while retaining over 95% of their initial efficiency after 1000 hours of continuous operation at maximum power point under 1-sun illumination in a N_2 atmosphere at 65 °C. Furthermore, 4APM-based perovskite solar modules (PSMs) with an active area of 14.0 cm^2 deliver a PCE of 23.16%. Our findings underscore the critical role of functional group spatial distribution in the rational design of molecular passivators for perovskite photovoltaics.

Introduction

Hybrid metal halide perovskite materials have emerged as exceptional light absorbers for next-generation photovoltaics, with certified power conversion efficiencies (PCEs) exceeding 27% over the past decade.^[1-4] However, solution-based fabrication of perovskite solar cells (PSCs) inevitably generates uncoordinated ion defects at perovskite film surfaces and grain boundaries, which critically undermines device stability.^[5-7] In-situ modification has emerged as a widely adopted strategy in conventional PSCs for suppressing non-radiative charge carrier recombination and enhancing device durability.^[8-9] By incorporating functional molecules directly into the perovskite precursor solution, this approach promotes carrier transport and modulates crystallization kinetics, enabling comprehensive passivation of bulk defects throughout the entire film.^[10-11]

Incorporating amino-based molecules with tailored functional groups as precursor additives has proven effective in modulating the crystallization kinetics of formamidinium (FA) and cesium (Cs)-based perovskites while efficiently passivating perovskite defects.^[12-14] Previous studies have demonstrated that variations in substituent chemistry significantly influence the passivation efficacy of amino-based molecules, with representative examples including long-chain alkyl groups^[15, 16], pyridyl groups^[17, 18], and aromatic groups^[19, 20]. Furthermore, increasing the number of amino functional groups within a single molecule has been shown to substantially enhance its overall passivation capability.^[21] Nevertheless, the role of the intramolecular spatial distribution of multiple amino groups in governing passivation performance remains poorly understood.^[22, 23] Therefore, a systematic investigation into the relationship between the intramolecular spatial arrangement of passivating groups and their molecular passivation capacity is critically needed.

In this work, we investigate the effect of amino group spatial distribution in molecular passivators on their passivation capability and PSC performance by introducing bis(4-aminophenyl)methane (2APM), tris(4-aminophenyl)methane (3APM), and tetrakis(4-aminophenyl)methane (4APM), into the perovskite precursor solution (Fig. 1a and Supplementary Fig. 1). Furthermore, another molecule, tetrakis(4-aminophenyl)ethylene (4APE), is also studied as a reference, which possesses the same number yet relatively more planar distribution of amino groups compared to 4APM.^[24] The study found that 4APM with its amino groups spatially distributed in a three-dimensional (3D) pattern, relative to the other three molecules, present more obvious enhancements in defect passivation, and crystallinity regulation. Meanwhile, the 4APM molecule can optimize the interfacial energy level alignment, facilitating efficient charge carrier extraction. As a result, 4APM-based PSCs and perovskite solar modules with an active area of 14.0 cm^2 achieve the best power conversion efficiencies (PCEs) of 26.26% and 23.16%, respectively, while retaining over 95% of their initial efficiency after 1000 hours of continuous operation under maximum power point conditions.

Results

Molecular interactions with perovskite

To investigate the effect of amino group spatial distribution on the passivation capability, density functional theory (DFT) calculations were conducted. As shown in Fig. 1b, the electron cloud density of the amino group increases from -0.195 for 2APM to -0.197 for 3APM and -0.212 for 4APM. Correspondingly, the molecular dipole moment follows the same trend, rising from 0.72 D for 2APM to 0.75 D for 3APM and 1.95 D for 4APM, which indicates that the 3D spatial configuration of amino groups plays a significant role in enhancing the molecular dipole moment, thereby strengthening the electrostatic interaction with perovskite surface and promoting charge redistribution at the molecule/perovskite interface.^[25]

To further quantify additive–perovskite interactions, the Vienna Ab Initio Simulation Package (VASP) was employed to calculate the interaction energies (E_{int}) between each additive and uncoordinated I vacancy (V_{I}) defects in the perovskite lattice.^[26, 27] As shown in Fig. 1c, Supplementary Fig. 2 and Supplementary Table 1, 4APM exhibits the strongest interaction energy of -0.791 eV among the three molecules. The formation energies of V_{I} defects before and after modification (Supplementary Fig. 3-4) shows that the 4APM-modified perovskite exhibits the higher value of 0.525 eV, compared to other counterparts, indicating highly difficult to form lead ion defects in the 4APM sample. Charge density difference plots further reveal that 4APM facilitates greater electron density sharing toward Pb^{2+} relative to the other additives (Fig. 1d and Supplementary Fig. 5). These DFT results demonstrate that extending the spatial distribution of amino groups progressively enhances the ability of amino-containing additives to suppress defect formation in perovskite films, thereby facilitating the growth of higher-quality perovskite films during annealing (Fig. 1e).^[28]

To further investigate the interactions between the three molecules and Pb^{2+} , proton nuclear magnetic resonance (^1H NMR) experiments were conducted (Fig. 2a-c). The chemical shifts of $-\text{NH}_2$ group in 2APM, 3APM and 4APM shifted upfield by 0.04 , 0.07 and 0.10 ppm upon addition of PbI_2 , respectively, indicating progressively stronger interaction with Pb^{2+} as the spatial distribution of amino groups increases. Notably, 4APM exhibited the largest upfield shift, confirming its strongest interaction with Pb^{2+} among the three additives.^[25] In addition, mixing these molecules with formamidinium iodide (FAI) also induced obvious chemical shifts in the proton signals of $-\text{NH}_2$ groups, suggesting non-negligible interactions between the amino groups and FAI (Supplementary Fig. 6).

The coordination behavior between these amino-based molecules and Pb^{2+} was further examined by Fourier Transform Infrared Spectroscopy (FTIR) measurements, where each additive was mixed with PbI_2 at a molar ratio of $-\text{NH}_2$ group to PbI_2 of 1:1 (Fig. 2d-f, and Supplementary Fig. 7). After mixing with PbI_2 , the N-H stretching and bending vibrational peaks of all three additives shifted to lower wavenumbers, confirming the coordination between amino groups and Pb^{2+} .^[29] In specific, the 4APM-based sample exhibited the more pronounced red-shifts (N–H bending: $1616\text{ cm}^{-1} \rightarrow 1600\text{ cm}^{-1}$, $\Delta = 16\text{ cm}^{-1}$; N–H stretching: $3429\text{ cm}^{-1} \rightarrow 3413\text{ cm}^{-1}$, $\Delta = 16\text{ cm}^{-1}$) than the other two counterparts. These results consistently demonstrate that 4APM with 3D distribution of amino groups possesses the strongest coordination ability with Pb^{2+} , in good agreement with the DFT and ^1H NMR findings.

X-ray photoelectron spectroscopy (XPS) measurements were performed on the additive-treated perovskite films to further elucidate the interactions between the additives and Pb^{2+} defects. As shown in Fig. 2g, the Pb 4f spectra exhibit binding-energy shifts of 0.08 eV for 2APM, 0.12 eV for 3APM and 0.16 eV for 4APM, respectively. The N 1s spectra follow the same trend (Fig. 2h), indicating that more stereoscopic spatial distribution of $-\text{NH}_2$ groups enhances the defect passivation efficacy.^[30] Furthermore, the I/Pb ratios of the perovskite films were determined by XPS (Supplementary Fig. 8). The control film exhibits a I/Pb ratio of 3.53 , whereas 2APM-, 3APM-, and 4APM-treated films display progressively higher ratios of 3.68 , 3.79 , and 3.91 , respectively. The increased I/Pb ratio in the 4APM-modified film implies more effective suppression of volatile I^0 species formation, thereby reducing the density of uncoordinated Pb^{2+} defects.

To decouple the influence of amino groups quantity from their spatial distribution, we introduced a more planar amino-based molecule 4APE as a structural control (Supplementary Fig. 9-11). Compared to 4APM, 4APE exhibits a denser electron cloud distribution yet a significantly lower dipole moment, despite containing the same number of amino groups. Both computational

and experimental results consistently demonstrate that while the number of amino groups contributes to the interaction energy, the 3D configuration of 4APM enables its four -NH_2 groups to extend more freely and uniformly into space, in favor of stronger and more spatially distributed interactions with perovskite defects. This stereoscopically structural advantage amplifies the molecular dipole moment of 4APM relative to the more planar 4APE, thereby improving its defect passivation capability beyond what the count of functional groups alone can achieve.

Molecular effects on film properties

The effect of 2APM, 3APM and 4APM on perovskite crystal growth and film morphology was further investigated. As shown in Fig. 2i and Supplementary Fig. 12, additive-treated perovskite films exhibit more intense (100) X-ray diffraction (XRD) peaks and narrower full-width at half-maximum (FWHM) values relative to the control sample. Notably, both peak intensity and FWHM follow a clear amino spatial distribution-dependent trend, with 4APM-treated perovskite film yielding the highest crystallinity among the three samples. To further corroborate these findings, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were conducted. Compared to the control, 2APM- and 3APM-treated perovskite films, the 4APM-treated sample exhibits enhanced diffraction intensity along the out-of-plane (q_z) direction at $q \approx 1 \text{ \AA}^{-1}$ (Supplementary Fig. 13 and 14), which corresponds to the (100) crystal plane, indicating stronger preferential orientation, in good agreement with the XRD results discussed above. These results confirm that the introduced additives enhance perovskite crystallinity, which becomes increasingly pronounced with the higher spatial distribution of amino groups. In addition, in-situ XRD measurements (Supplementary Fig. 15) reveal a consistent trend, wherein the additive, particularly 4APM, retards perovskite crystallization kinetics more effectively, promoting the formation of larger perovskite grains and higher-quality films. [33]

The surface morphologies of the four perovskite films were examined by scanning electron microscopy (SEM), as shown in Fig. 3a-d. The results clearly show that additive incorporation, particularly the 4APM molecule, significantly promotes perovskite grain growth, yielding an average grain size of $1.95 \text{ }\mu\text{m}$, higher than those of other samples (Supplementary Fig. 16). In addition, cross-sectional SEM measurements further reveal that the 4APM-treated perovskite film exhibits a more continuous grain structure with fewer and less distinct grain boundaries compared to control and those treated with 2APM and 3APM (Supplementary Fig. 17). Atomic force microscopy (AFM) analysis (Supplementary Fig. 18) also reveals a progressive reduction in the surface roughness upon the treatment of 2APM, 3APM, and 4APM. The smoother and more uniform surfaces achieved, most prominently with 4APM, are expected to improve interfacial contact and minimize carrier transport losses, thereby facilitating more efficient charge extraction. [34]

Two-dimensional (2D) photoluminescence (PL) mapping was conducted to spatially visualize how the spatial distribution of amino groups influences defect passivation capability across the perovskite films. As shown in Fig. 3e-h, the introduction of additives leads to notable improvements in both PL intensity and spatial uniformity, with 4APM-treated films exhibiting the most homogeneous emission distribution. [35] These observations underscore the critical role of more extended distribution of amino groups in mitigating defect states and suppressing nonradiative recombination in perovskite films.

The influence of the additives on charge carrier dynamics was further assessed by steady-state and time-resolved photoluminescence (TRPL) measurements. As illustrated in Fig. 3i, 4APM-modified perovskite device exhibits a remarkable enhancement in PL emission intensity relative to the other devices, indicating most effectively reduced non-radiative recombination. The TRPL lifetime fitting was conducted using the double exponential method. The fast lifetime (τ_1) represents non radiative recombination on the surface/interface, while the slow lifetime (τ_2) corresponds to radiative recombination in the bulk phase, as shown in Fig. 3j and Supplementary Table 2. [36] The carrier lifetimes increase with the increased spatial distribution of amino groups, confirming progressive suppression of trap-assisted recombination.

The photoluminescence quantum yield (PLQY) of the corresponding full devices was determined via absolute intensity photoluminescence (AIPL) measurements to further quantify the effect of the spatial distribution of amino groups on film optoelectronic quality. As shown in Fig. 3k, the 4APM-treated device exhibits a markedly higher average PLQY of 5.89%,

compared to 2.83% for 3APM, 2.44% for 2APM, and 1.72% for the control, reflecting significant reduction of nonradiative recombination in 4APM-modified device.^[37] The quasi-Fermi level splitting (QFLS) was calculated from the PLQY using the equation (Supplementary Note 1).^[38] The extracted QFLS/q values increase progressively from 1.166 V for the control to 1.179 V, 1.189 V, and 1.220 V for 2APM-, 3APM-, and 4APM-modified devices, respectively, confirming that 4APM most effectively suppresses non-radiative recombination and elevates the intrinsic optoelectronic quality of the perovskite films.

Ultraviolet photoelectron spectroscopy (UPS) measurements were conducted to investigate the influence of additives on the energy levels of perovskite layer (Supplementary Fig. 19-20). The valence band of the 4APM-treated film exhibits positive shifts of 40 meV, 100 meV, and 150 meV compared to 3APM-, 2APM-treated and the control films, respectively, resulting in improved energy level alignment with the HTL layer (Fig. 3l).^[39] These results demonstrate that additive incorporation, especially 4APM, simultaneously passivates defects, enhances carrier lifetime, and optimizes interfacial energy level alignment, contributing to reduced non-radiative recombination and improved charge transport throughout the device.^[40]

Conductive AFM was further employed to verify the effect of the additives on local charge transport within the perovskite films (Fig. 4a-d). Upon additive incorporation, the average current increases systematically with the spatial distribution of amino groups, from 75.1 pA for the control to 105.4 pA for 2APM, 146.2 pA for 3APM, and 189.7 pA for 4APM, respectively, confirming progressively enhanced film conductivity. To further validate the additive-induced changes in the surface electronic structure, Kelvin probe force microscopy (KPFM) was employed to map the surface potential distribution of the perovskite films (Fig.s 4e-h). Notably, the large surface potential fluctuations associated with grain boundaries in the untreated film are nearly eliminated after 4APM treatment, yielding a highly uniform surface potential distribution. This improvement can be attributed to the surface dipoles introduced by 4APM, which suppress potential fluctuations and reduce surface defect density.^[41-44] Furthermore, the contact potential difference (CPD) decreases progressively with increasing the spatial distribution of amino groups, from 189.3 mV for the control to 126.5 mV, 83.4 mV, and 37.7 mV for 2APM-, 3APM-, and 4APM-modified films (Supplementary Fig. 21), respectively, reflecting an increasingly favorable surface electronic environment for charge extraction.

Time-of-flight secondary ion mass spectrometry (TOF-SIMS) measurements were conducted to investigate the distribution of the additives within the perovskite films. Depth-profile analysis reveals that the additive molecules are partially enriched at the upper surface of the perovskite layer (Supplementary Fig. 22), owing to the synergistic effects of physical segregation during perovskite crystallization and molecular chemical affinity for surface defects, which favors effective passivation of under-coordinated Pb^{2+} defects at the surface and grain boundaries of perovskite films.^[45,46] Atomic Force Microscopy-based Infrared Spectroscopy (AFM-IR) was further employed to examine the surface properties of the perovskite thin films. Infrared imaging at 1519 cm^{-1} , 1508 cm^{-1} , and 1502 cm^{-1} , corresponding to the characteristic aromatic C-H bending signals of 2APM, 3APM, and 4APM, respectively (Supplementary Fig. 23), reveals that the additive-treated perovskite films exhibit significantly stronger and more spatially homogeneous benzene ring signals compared to the control (Fig. 4i-l), confirming uniform distribution of the molecular additives across the perovskite surface. These results demonstrate that the incorporated additives, particularly 4APM, play a crucial role in promoting perovskite crystallization, suppressing defect formation and improving charge transport throughout the film.^[47]

Photovoltaic performance and device stability

To investigate the influence of molecular additives on device performance, PSCs with the architecture FTO/C-TiO₂/SnO₂/perovskite/Spiro-MeOTAD/Ag were fabricated (Fig. 5a). Current density-voltage (J - V) curves measured across a range of additive concentrations are shown in Supplementary Fig. 24, with the corresponding photovoltaic (PV) parameters summarized in Supplementary Tables 3-5. The optimal additive concentration was determined to be 0.1 mg/mL for all three molecules. The J - V curves of the champion devices under AM 1.5G illumination are presented in Fig. 5b, and the key PV data are listed in Table 1. The control device yields a power conversion efficiency (PCE) of 24.04%, with an open-circuit voltage (V_{oc}) of 1.155 V, a short-circuit current density (J_{sc}) of 25.42 mA cm^{-2} , and a fill factor (FF) of 81.87%. Incorporation of additives 2APM and 3APM enhances the device performance, achieving PCEs of 24.74% and 25.67%, respectively. Notably, 4APM-based device delivers the

highest PCE of 26.26%, with a V_{OC} of 1.187 V, J_{SC} of 25.72 mA cm⁻², and FF of 85.95%. The progressive enhancement in V_{OC} across the additive-treated devices can be attributed to the synergistic effects of suppressed non-radiative recombination and optimized interfacial energy level alignment.^[48] Under steady-state maximum power point tracking over 500 seconds, the 4APM-based device maintains a stabilized PCE of 26.04%, the highest among all tested devices (Fig. 5c). Furthermore, the 4APM-modified device exhibits a negligible hysteresis of 0.7%, compared to 1.2% for 3APM, 1.8% for 2APM, and 2.3% for the control (Supplementary Fig. 25 and Supplementary Table 6). Perovskite solar modules (PSMs) with an active area of 14.0 cm² were further fabricated to assess scalability, the J - V curves of the PSMs with different additives are shown the Fig. 5d. The control PSM shows a PCE of 19.30%, with a V_{OC} of 10.77 V, J_{SC} of 2.376 mA cm⁻², and FF of 75.61%. While 2APM- and 3APM-treated PSMs achieve higher PCEs of 20.52% and 22.05%, respectively. Remarkably, the 4APM-modified PSM delivers the highest efficiency of 23.16%, with a V_{OC} of 11.56V, J_{SC} of 2.406 mA cm⁻², and FF of 83.23% (Supplementary Table 7), demonstrating the effective scalability of the modification strategy with the 3D amino distribution.

Incident photon-to-electron conversion efficiency (IPCE) spectra of all devices are shown in Fig. 5e. The integrated current densities of the control, 2APM-, 3APM-, and 4APM-treated devices are 24.51, 24.65, 24.88, and 25.17 mA cm⁻², respectively, which are consistent with the J_{SC} values obtained from J - V measurements (Table 1). The improved photovoltaic performance of 4APM-based devices was further corroborated by statistical analysis of the PV parameters across 20 independently fabricated devices (Fig. 5f, Supplementary Fig. 26 and Supplementary Table 8). Consistently, the ideality factors extracted from voltage response fitting decrease systematically with increasing the spatial distribution of amino groups, from 1.82 for the control to 1.67, 1.48, and 1.37 for 2APM-, 3APM-, and 4APM-based devices (Fig. 5g), respectively, confirming progressively reduced trap-assisted nonradiative recombination.^[49] The trap state density in perovskite films was quantitatively assessed using the space charge limited current (SCLC) method. As shown in Supplementary Fig. 27, the pristine film exhibits a trap state density of 9.39×10^{15} cm⁻³, which decreases progressively to 7.12×10^{15} cm⁻³, 5.05×10^{15} cm⁻³, and 3.71×10^{15} cm⁻³ upon treatment with 2APM, 3APM, and 4APM, respectively, demonstrating the best defect passivation capability of 4APM.

To further investigate the impact of amino distribution on film stability, XRD measurements were conducted to monitor the phase stability of perovskite films with and without additives upon exposure to ambient air at ~35% relative humidity (RH). As shown in Supplementary Fig. 28, the control film exhibits a weak δ -FAPbI₃ peak after 1000 h, which significantly intensifies after 2000 h. 2APM- and 3APM-treated films show negligible δ -FAPbI₃ and PbI₂ peaks after 1000 h, which, however, intensify upon further aging to 2000 h. In contrast, the 4APM-incorporated film demonstrates markedly superior phase stability, maintaining a negligible δ -FAPbI₃ peak even after 2000 h of exposure, confirming the exceptional moisture-resistance conferred by the 3D amino group distribution. Furthermore, long-term storage stability of unencapsulated devices was evaluated under 35% RH. The 4APM-modified PSCs exhibit the best durability, retaining 93% of their initial PCE after 2000 h of aging (Fig. 5h), substantially outperforming the 2APM- (85%) and 3APM-based (88%) devices over the same period, and the control device, which retains only 38% of its initial PCE after 1450 hours. Under maximum power point (MPP) tracking conditions at 65 °C in a nitrogen (N₂) atmosphere (Fig. 5i), the unencapsulated 4APM-based device maintains 95% of its initial efficiency after 1000 h under continuous illumination, while the control, 2APM-, and 3APM-modified devices retain only 52%, 77%, and 86% of their initial PCEs, respectively. Furthermore, we investigated the operational stability of the encapsulated PSM under continuous light illumination at 65 °C in ambient air (Supplementary Fig. 29). After 600 hours, the 4APM-treated PSM retained 92% of its initial PCE, whereas the control device lost 30% of its initial PCE, further underscoring the superior long-term operational stability imparted by the modification strategy based on the 3D distribution of passivation groups.

Discussion

In summary, we have systematically investigated the effect of amino group spatial distribution in molecular passivators on photovoltaic performances. Integrating DFT calculations with comprehensive experimental characterizations, we demonstrate that increasing the spatial distribution of amino groups progressively enhances defect passivation efficacy, suppresses non-radiative

recombination, facilitates interfacial charge transport, and improves perovskite film crystallinity, driving systematic improvements in device performances. The optimized PSC incorporating 4APM with the 3D distribution of amino group achieves a stabilized PCE of 26.26%, while the corresponding PSM with an active area of 14.0 cm² delivers a PCE of 23.16%. Furthermore, 4APM-based unencapsulated devices retain 95% of their initial PCE after 1000 hours of continuous MPP operation at 65 °C in a N₂ atmosphere, and 93% after 2000 hours of dark storage under ambient conditions, demonstrating exceptional long-term operational and environmental stability. This work establishes the spatial distribution of functional groups as a critical design parameter for molecular additives, offering a clear and generalizable strategy for the development of high-efficiency and stable perovskite photovoltaics.

Methods

Materials: Formamidineum iodide (FAI), methylammonium lead bromide (MAPbBr₃), cesium iodide (CsI), methylammonium chloride (MACl), lithium bis(trifluoromethanesulfonyl) imide (Li-TFSI), 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (Spiro-OMeTAD) (99.8%), Poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA), 4-tert-butylpyridine and lead iodide (PbI₂), 4-tert-butylpyridine (tBP), SnCl₄ were purchased from Xi'an Polymer Light Technology Corp in China. SnO₂ colloid precursor (tin (IV) oxide) was purchased from Alfa Aesar. N, N-Dimethylformamide (DMF), dimethyl sulfoxide (DMSO), isopropyl alcohol (IPA), chlorobenzene (CB), ethyl acetate (EA), acetonitrile (ACN), bis(4-aminophenyl)methane (2APM), tris(4-aminophenyl)methane (3APM), tetrakis(4-aminophenyl)methane (4APM) and tetrakis(4-aminophenyl)ethylene (4APE) were purchased from Shanghai Macklin Biochemical Co., LTD. All chemicals were used as received without further purification.

Device Fabrication: The FTO-coated glass was cleaned via sequential sonication (30 min for each) with detergent once, deionized water twice, and ethanol for twice. Then the cleaned substrate was treated with UV/O₃. A compact layer of TiO₂ was deposited by atomic layer deposition (ALD) and annealed in ambient air at 500 °C for 30 min. Anhydrous SnCl₄ and deionized water were mixed in a volume ratio of 1: 1.75 and aged at 25 °C for a week, and the solution changed from a clear and transparent solution to a white colloidal solution (SnO₂). The SnO₂ electron transport layer was deposited by spin-coating at 3,000 rpm for 30 s, and annealed at 180 °C for 30 min. The perovskite solution ((FA_{0.95}Cs_{0.05}) PbI₃)_{0.975}(MAPbBr₃)_{0.025}) was prepared by mixing 705.3 mg of PbI₂, 228.8 mg of FAI, 18.2 mg of CsI, 33.67 mg of MACl and 18.2 mg of MAPbBr₃ in mixed DMF/DMSO solvent system (v: v=9:1). The amounts of bis(4-aminophenyl)methane (2APM), tris(4-aminophenyl)methane (3APM), and tetrakis(4-aminophenyl)methane (4APM) added were from 0.1-0.5 mg/mL. The perovskite solution was spin-coated on the SnO₂ substrates in a two-step at 1000 rpm/10 s and 4000rpm/30 s. In the last 15 s of the second step, 300 µl of the anti-solvent anisole was dropped at a constant rate, after spin coating was completed, the perovskite film was annealed at 100 °C for 40 min. Then the hole transport layer was prepared by mixing Spiro-OMeTAD (72.3 mg), 4-tert-butylpyridine (28.8 µl), and Li-TFSI (17.5 µl) solution (520 mg of Li-TFSI in 1 ml of acetonitrile) in chlorobenzene (1 ml). The prepared Spiro-OMeTAD solution (40 µl) was deposited on perovskite films by spin-coating (3000 rpm, 30s). Finally, 80 nm thick Ag electrodes were thermally evaporated under vacuum to complete the device fabrication. For devices in 85° thermal stability tests, Spiro-OMeTAD was replaced with PTAA (PTAA (20 mg), preparation of PTAA solution by mixing 4-tert-butylpyridine (10 µl) and Li-TFSI (15 µl) solution (170 mg Li-TFSI in 1 mL acetonitrile) in chlorobenzene (1 mL)). Finally, 80 nm thick Au electrodes were thermally evaporated under vacuum to complete the device fabrication.

Module fabrication: P1 (19 µm wide) lines etching process was pre-patterned on FTO glass (5 cm × 5 cm) with a 1064 nm fiber laser (Han's laser). The laser power ratio, laser duty cycle, and laser frequency were 30%, 5%, and 50 kHz, respectively. Then, patterned FTO substrates were cleaned and treated by UV Ozone Cleaner (Ossila) for 15 min. The SnO₂, Perovskite, and Spiro-OMeTAD layers were prepared with the same procedure as presented. For the 160-µm-wide P2 lines etching process, the laser used was a 532 nm laser with a laser power ratio of 65%, a laser duty cycle of 5%, and a laser frequency of 100 kHz. 80 nm thick

Au electrodes were thermally evaporated under a vacuum to complete the fabrication of the modules. Finally, 78- μm -wide P3 lines etching used the same laser as P2 with a laser power ratio of 50%, a laser duty cycle of 5%, and a laser frequency of 100 kHz. P4 is an etching procedure for cleaning the edge of the modules, the laser used in P4 is the same as P1 with a laser power ratio of 40%, a laser duty cycle of 10%, and a laser frequency of 100 kHz.

Computational details: All the calculations are performed using the Gaussian 16W software package. The geometry optimization and frequency calculation are carried out using the B3LYP functional with the LANL2DZ basis set. The structure is fully optimized, and a frequency calculation is performed to ensure that the optimized geometry corresponds to a true minimum (no imaginary frequencies). The single-point energy calculations are also carried out at the same level of theory. The charge distribution is analyzed based on the population analysis method, and the electrostatic potential is calculated and visualized to examine the distribution of the charge in the molecular system. The convergence criterion for the geometry optimization is set to 10^{-4} a.u. for the energy change, and the maximum force is set to 0.00045 a.u. to ensure accurate results. In the Gaussian input file, fix the position of molecules using the NoOpt keyword to prevent geometric optimization during the calculation process.

All DFT calculations are implemented using the Vienna ab initio Simulation Package (VASP). We calculated the interaction energies of three molecules on the FAPbI_3 (100) surface and introduced I vacancies to create undercoordinated Pb^{2+} sites, simulating the typical dominant trap states. The electronic exchange-correlation interactions were described using the Perdew-Burke-Ernzerhof (PBE) functional method under the generalized gradient approximation (GGA-PBE). In all calculations, the plane-wave cutoff energy for electronic wavefunctions was set to 400 eV. The Brillouin zone sampling employed a $(2 \times 2 \times 1)$ Monkhorst-Pack k-point mesh for geometric structure relaxation. To avoid interactions between adjacent layers, a vacuum layer of 20 Å was set along the z-axis. The dipole moment correction function remained enabled. The van der Waals forces in the structure were corrected using Grimme's DFT-D3 method.

Materials Characterization: Scanning electron microscopy (SEM, S-4800, Hitachi) and atomic force microscopy (AFM, Bruker dimension Fast Scan) with peak-force tapping mode using silicon tips (OTESPA, Bruker) were used to characterize the morphology of the perovskite films. Fourier transform infrared (FTIR) spectroscopy was obtained from the instrument Thermo, Nicolet 6700. The resolution of the instrument is 4 cm^{-1} . The UV-Vis absorption spectra were measured by a U4100 spectrophotometer (Hitachi). X-ray diffraction (XRD) was performed in regular θ -2 θ scanning mode using X-ray diffractometer (PANalytical, $\text{CuK}\alpha_1$: $\lambda=0.154056 \text{ nm}$) with scan rate $10^\circ \text{ min}^{-1}$. The photoluminescence (PL) signal was recorded using a Horiba Jobin Yvon system, while the time-resolved PL decay profiles were measured using a PicoHarp 300 with time-correlated single-photon counting capabilities. Kelvin probe force microscopy (KPFM) was implemented based on a Bruker Dimension V SPM system. The equipped AFM tip was Pt/Ir coated, with $\sim 60 \text{ kHz}$ resonance frequency, purchased from Bruker. Ultraviolet photoelectron spectroscopy (UPS) was performed on a photoelectron spectrometer (ESCALAB 250Xi, Thermo Fisher Scientific). The XPS test was carried out with a photoelectron spectrometer model of ESCA LAB 250Xi from Thermo Fisher Scientific in the United States. Compositional depth profiling of perovskite films was carried out using a TOF-SIMS 5 system from IONTOF, which was operated in the spectral mode and using a 30-keV Bi^{3+} primary ion beam with an ion current of 0.8 pA. Photoluminescence quantum yield (PLQY) was measured using an integrating sphere (Fluorolog, Horiba JobinYvon), an Andor Kymera 193i spectrograph, and a 660 nm continuous-wave laser (OBIS, Coherent) set at 1-Sun equivalent photon flux ($1.1 \mu\text{m}$ beam full-width half-maximum, $632 \mu\text{W}$); photoluminescence was collected at normal incidence using a 0.1 NA, 110 μm -diameter optical fiber. For the absolute spectral calibration of the PLQY measurement system, we used a radiometrically calibrated halogen lamp (HL-3 plus CAL from Ocean Optics). PL mapping was measured with a laser scanning confocal microscope (Enlitech, SPCM-1000) equipped with a 470-pulse laser and galvo-based scanner. The objective was $\times 50$ with numerical aperture of 0.8; for this objective, the confocal pinhole was set to 50 μm size. The detector was a photo-multiplier tube (Hamamatsu) with a photon-counting module.

Device Characterization: A 2400 Source Meter (Keithley, USA) was used to measure the J - V curve of the device at a simulated AM 1.5 G 100 mW/cm^2 intensity (Oriel Sol3A Class AAA, Newport, USA) with a scan rate 0.1 V/s (1.3 V to -0.1 V). The cells

with an active area of 0.25 cm² (0.5 cm × 0.5 cm) were masked with a black metal mask with an area of 0.09 cm². The external quantum efficiency (EQE) measurement was recorded using certified incident photon to current conversion efficiency equipment from Enlitech (Taiwan). The MPP stability tests were performed using a stability setup (LC Auto-Test 24, Shenzhen Lancheng Technology Co., Ltd.). In the test, all PSCs were without encapsulation, and tested under continuous light illumination and maximum power point tracking. The light source consisted of an array of white LEDs powered by a constant current. The LED type is MG-A200A-AE with an emission spectrum of 400-750 nm. Equivalent sun intensities were calibrated using a calibrated Si-reference cell. During aging, the PSCs were masked and placed inside a sample holder purged with continuous N₂ flow. *J-V* curves with reverse voltage scans were recorded every 6 h during the whole operational test.

Data Availability

The data that support the findings of this study are available within the article and its supplementary information. Correspondence and requests for materials should be addressed to the corresponding authors (Z.S., C.L., M.K.N. and X.G.).

References

1. Zhao, K., *et al.* Peri-Fused polyaromatic molecular contacts for perovskite solar cells. *Nature* **632**, 301-306 (2024).
2. Yang, Y., *et al.* Hafnium oxide interface stabilization for efficient, photothermally stable perovskite solar cells. *Science* **391**, 926-930 (2026).
3. Tian, C., *et al.* Photostable donor–acceptor interface for minimizing energy loss in inverted perovskite solar cells. *Nat. Photonics* **20**, 1-9 (2026).
4. Xiong, Z., *et al.* Homogenized chlorine distribution for >27% power conversion efficiency in perovskite solar cells. *Science* **390**, 638-642 (2025).
5. Zhang, T., *et al.* On the role of molecular conformation of the passivating agents for high-performance perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202505667 (2025).
6. Wang, C., *et al.* A universal close-space annealing strategy towards high-quality perovskite absorbers enabling efficient all-perovskite tandem solar cells. *Nat. Energy* **7**, 744 (2022).
7. Wu, J., *et al.* Regioselective multisite atomic-chlorine passivation enables efficient and stable perovskite solar cells. *J. Am. Chem. Soc.* **145**, 5872-5879 (2023).
8. Zhang, Q., *et al.* Multifunctional butylammonium acetate additive regulates FAPbI₃ film growth and defect passivation for highly efficient and stable perovskite solar cells with 26.12% efficiency. *Adv. Funct. Mater.* **35**, 2507934 (2025).
9. Jiang, R., *et al.* Crystallization modulation for wide-bandgap perovskites with universal defect passivation toward efficient perovskite/organic tandem photovoltaics. *Nat Commun* **17**, 1375 (2026).
10. Shen, L., *et al.* Ion-diffusion management enables all-interface defect passivation of perovskite solar cells. *Adv. Mater.* **35**, 2301624 (2023).
11. Duong, N., *et al.* The reactive nature of formamidineium: Amine passivation induces heterostructure formation in inverted perovskite cells. *Joule* **9**, 102055 (2025).
12. Zhou, N., *et al.* Homologous molecule treatment in perovskite solar cells: synergistic management of holistic defect and charge transfer. *Adv. Funct. Mater.* **35**, 2418798 (2025).
13. Yang, T., *et al.* One-stone-for-two-birds strategy to attain beyond 25% perovskite solar cells. *Nat Commun* **14**, 839 (2023).
14. Geng, S., *et al.* Revealing collaborative effects of binary additives on regulating precursor crystallization toward highly efficient perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202424910 (2025).
15. Wu, W., *et al.* Bilateral alkylamine for suppressing charge recombination and improving stability in blade-coated perovskite solar cells. *Sci. Adv.* **5**, 8925 (2019).
16. Zheng, X., *et al.* Managing grains and interfaces via ligand anchoring enables 22.3%-efficiency inverted perovskite solar cells. *Nature Energy*

- 5, 131–140 (2020).
17. Sun, X., *et al.* Vapor-assisted surface reconstruction enables outdoor-stable perovskite solar modules. *Science* **388**, 957–963 (2025).
18. Li, C., *et al.* Synergistic electron-deficient surface engineering: A key factor in dictating electron carrier extraction for perovskite photovoltaics. *J. Am. Chem. Soc.* **147**, 25738–25749 (2025).
19. Yun, Y., *et al.* Dimensional engineering of interlayer for efficient large- area perovskite solar cells with high stability under ISOS- L- 3 aging test. *Sci. Adv.* **11**, 3112 (2025).
20. Zhang, T., *et al.* On the role of molecular conformation of the passivating agents for high-performance perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202505667 (2025).
21. Su, L., *et al.* Performance enhancement of perovskite solar cells using trimesic acid additive in the two-step solution method. *J. Power Sources* **426**, 11–15 (2019).
22. Wu, J., *et al.* Regioselective multisite atomic chlorine passivation enables efficient and stable perovskite solar cells. *J. Am. Chem. Soc.* **145**, 5872–5879 (2023).
23. Liu, C., *et al.* Dimensional regulation of organic n-type dopants for highly efficient perovskite solar cells and modules. *Adv. Mater.* **37**, 2417251 (2025).
24. Liu, J., *et al.* Through-space interaction of tetraphenylethylene: what, where, and how. *J. Am. Chem. Soc.* **144**, 7901–7910 (2022).
25. Zhang, H., *et al.* Fluorine-containing passivation layer via surface chelation for inorganic perovskite solar cells. *Angew. Chem. Int. Ed.* **62**, e202216634 (2022).
26. Li, T., *et al.* Inorganic wide-bandgap perovskite subcells with dipole bridge for all-perovskite tandems. *Nature Energy* **8**, 610–620 (2023).
27. Gu, X., *et al.* Rational surface-defect control via designed passivation for high-efficiency inorganic perovskite solar cells. *Angew. Chem. Int. Ed.* **60**, 23164–23170 (2021).
28. Zhou, Q., *et al.* Stronger binding force improving surface passivation of perovskites for High-Performance inverted solar cells. *Chem. Eng. J.* **440**, 135974 (2022).
29. Dai, R., *et al.* Pre-buried Interface Strategy for Stable Inverted Perovskite Solar Cells Based on Ordered Nucleation Crystallization. *Adv. Funct. Mater.* **33**, 2305013 (2023).
30. Liu, L., *et al.* Collaborative strategy of multifunctional groups in trifluoroacetamide achieving efficient and stable perovskite solar cells. *Sol. RRL* **6**, 2200284 (2022).
31. Yang, T., *et al.* Tailoring the interfacial termination via dipole interlayer for high-efficiency perovskite solar cells. *Adv. Energy Mater.* **13**, 2204192 (2023).
32. Duan, J., *et al.* Effect of side-group-regulated dipolar passivating molecules on CsPbBr₃ perovskite solar cells. *ACS Energy Lett.* **6**, 2336–2342 (2021).
33. Yang, G., *et al.* Realizing uniform defect passivation via self-polymerization of benzenesulfonate molecules in perovskite photovoltaics. *Adv. Mater.* **37**, 2503435 (2025).
34. Chai, W., *et al.* Graded heterojunction improves wide-bandgap perovskite for highly efficient 4-terminal perovskite/silicon tandem solar cells. *Research* **6**, 0196 (2023).
35. Wang, K., *et al.* Ion-dipole interaction enabling highly efficient CsPbI₃ perovskite indoor photovoltaics. *Adv. Mater.* **35**, 2210106 (2023).
36. Zhou, B., *et al.* Strain engineering and halogen compensation of buried interface in polycrystalline halide perovskites. *Research*, **7**, 0309 (2024).
37. Kirchartz, T., *et al.* Photoluminescence-based characterization of halide perovskites for photovoltaics. *Adv. Energy Mater.* **10**, 1904134 (2020).
38. Caprioglio, P., *et al.* On the relation between the open-circuit voltage and quasi-fermi level splitting in efficient perovskite solar cells. *Adv. Energy Mater.* **9**, 1901631 (2019).
39. Xie, L., *et al.* Multifunctional anchoring of Oligands for high-performance and stable inverted perovskite solar cells. *InfoMat* **5**, e12379 (2023).

40. Lanzoni, E. M., *et al.* The impact of Kelvin probe force microscopy operation modes and environment on grain boundary band bending in perovskite and Cu(In, Ga)Se₂ solar cells. *Nano Energy* **88**, 106270 (2021).
41. Lin, X., *et al.* Dipole-field-assisted charge extraction in metal-perovskite-metal back-contact solar cells. *Nat Commun* **8**, 613 (2017).
42. Shao, W., *et al.* Tailoring perovskite surface potential and chelation advances efficient solar cells. *Adv. Mater.* **36**, 2310080 (2024).
43. Stecker, C., *et al.* Surface defect dynamics in organic–inorganic hybrid perovskites: from mechanism to interfacial properties. *ACS Nano* **13**, 12127 (2019).
44. Jiang, Q., *et al.* Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278 (2022).
45. Ding, Y., *et al.* Synergistic bulk and interface passivation via conjugated ionic additives enables 26% efficient PTAA-based perovskite solar cells. *Adv. Energy Mater.* e04647 (2025).
46. Luo, H., *et al.* Damp-stable perovskite/silicon tandem solar cells with internal encapsulating sulfonium-based molecules. *ACS Energy Lett.* **10**, 3325–3334 (2025).
47. Feng, Z., *et al.* Multi-site lead passivation via spatial configuration modulation of additives for efficient perovskite solar cells. *Adv. Energy Mater.* **15**, 2502409 (2025).
48. You, S., *et al.* Bifunctional hole-shuttle molecule for improved interfacial energy level alignment and defect passivation in perovskite solar cells. *Nat. Energy* **8**, 515–525 (2023).
49. Tress, W., *et al.* Interpretation and evolution of open-circuit voltage, recombination, ideality factor and subgap defect states during reversible light-soaking and irreversible degradation of perovskite solar cells. *Energy. Environ. Sci.* **11**, 151-165 (2018).

Acknowledgements

We acknowledge the Strategic Priority Research Program of the Chinese Academy of Sciences and the National Natural Science Foundation of China (NSFC) for their support. We thank the characterization of the Beamline 14B1 in the Shanghai Synchrotron Radiation Facility (SSRF) for grazing incident wide angle X-ray scattering (GIWAXS).

Funding Statement

This work was financially supported by the Strategic Priority Research Program of the Chinese Academy of Sciences (XDB1140000), the Natural Science Foundation of China (22209090), the Shandong Provincial Natural Science Foundation Project (ZR2022QB098), Shandong Energy Group Co., Ltd. Key R&D Program of Shandong Province (2024SFGC0101), Shandong Province Higher Education Institutions Youth Innovation Team Development Program (2023KJC016). Z.S. and M.F. acknowledge funding support from the University Research Fellowship (URF/R1/191286) and UKRI EPSRC Standard Grant (EP/WO25280/1).

Author Contributions Statement

X.J., G.Y. and B.Z. contributed equally. X.G., M.K.N., C.L. and Z.S. conceived the idea and supervised the research. X.J., G.Y. and B.Z. designed the experiments, fabricated PSCs and participated in sample preparation and characterization. Y.L. and P.W. contributed to DFT calculations and molecular dynamics simulations. N.S., K.D., L.Z., Y.Q., S.L., M.F. and S.P. coordinated the project and provided guidance and expertise. G.Y., X.J. and Z.S. wrote the manuscript draft. All authors participated in results discussions and manuscript review.

Competing Interests

The authors declare that they have no competing interests.

Tables

Table 1. Photovoltaic parameters of PSCs without and with 2APM, 3APM and 4APM under 1 sun illumination.

Sample	V_{OC} (V)	J_{SC} (mA cm ⁻²)	FF (%)	PCE (%)
--------	--------------	---------------------------------	--------	---------

Control	1.155	25.42	81.87	24.04
with 2APM	1.168	25.55	82.91	24.74
with 3APM	1.178	25.67	84.88	25.67
With 4APM	1.187	25.72	85.95	26.26

Figure Legends/Captions (for main text figures)

Fig. 1. Simulation of molecular characteristics and interactions with perovskite defects. (a) Molecular structures (ball-and-stick models) of bis(4-aminophenyl)methane (2APM), tris(4-aminophenyl)methane (3APM) and tetrakis(4-aminophenyl)methane (4APM) with 1D, 2D and 3D distributions of amino groups. (b) Calculated ESP profiles for 2APM, 3APM and 4APM molecules. (c) Interaction energies values of V_i defect sites within perovskite films with different additives. (d) Charge difference plot of 2APM, 3APM and 4APM on perovskite surfaces interacting with undercoordinated Pb^{2+} . (e) Schematic diagram of the chemical mechanism of using ammo-based molecules to regulate perovskite precursors, film crystallization, and defect passivation.

Fig. 2. Molecular interactions between additives and perovskite defects. (a-c) Proton nuclear magnetic resonance (1H NMR) spectra of 2APM, 3APM and 4APM with and without PbI_2 . (d-f) Fourier transform infrared spectroscopy (FTIR) spectra of 2APM, 3APM and 4APM films with and without PbI_2 . (g) Pb 4f XPS spectra, (h) N 1s XPS spectra, and (i) XRD patterns of perovskite films without and with 2APM, 3APM and 4APM treatment. (All figures use arbitrary units (arb. units) to denote relative intensity.)

Fig. 3. Molecular effects on perovskite film properties. (a-d) Top-view SEM images (scale bar: 1 μm), (e-h) two-dimensional PL intensity images, (i) Absolute intensity PL spectra of the corresponding devices (solid lines : averaged across three samples for each condition), and (j) TRPL decay of the perovskite films without and with 2APM, 3APM and 4APM. (k) Average PLQYs and quasi-fermi level splitting QFLS/q of the corresponding devices, averaged across three samples for each condition. Error bars represent the standard deviation. (l) Schematic diagrams of energy levels of perovskite films determined from UPS measurements. E_F is the Fermi level, and E_{VAC} is the vacuum level.

Fig. 4. Electrical characteristics of perovskite films. (a-d) Conductive AFM current mapping images (I_{avg} represents the average current across the images, scale bar: 1 μm), (e-h) KPFM images (scale bar: 1 μm), and (i-l) AFM-IR (benzene signals corresponding to additives at 1519 cm^{-1} (2APM), 1508 cm^{-1} (3APM), 1502 cm^{-1} (4APM), scale bar: 2 μm) images of Control, 2APM-, 3APM- and 4APM-treated perovskite films.

Fig. 5. Photovoltaic performance and device stability. (a) An illustration of the device architecture of PSCs. (b) Reverse $J-V$ curves of the PSCs without and with 2APM, 3APM and 4APM. (c) Time-dependent PCE of the PSCs measured at the maximum power point. (d) $J-V$ curves of the perovskite solar modules without and with 2APM, 3APM and 4APM. Inset: Image of the PSM. (e) IPCE spectra and corresponding integrated short-circuit current density (J_{sc}) of the PSCs. (f) Histogram of average PCE values of 20 devices without and with 2APM, 3APM and 4APM. (g) The light-intensity dependence of V_{oc} for PSCs. The ideality factor is calculated from the slope of the linear fit of the semilogarithmic plot; K , Boltzmann constant; T , Kelvin temperature; q , electron charge. (h) PCE evolution of the unencapsulated devices stored in dark and ambient environments (35% RH) for 2000 hours. Error bars represent the standard deviation from 20 independent measurements. (i) MPPT of the unencapsulated devices under simulated illumination at 65 $^{\circ}C$ in N_2 . (the test based on the device structure of FTO/C-TiO₂/SnO₂/perovskite/PTAA/Au).









